

tests/regression/ log_dir_writable_test.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Active standing regression guard (§11.4.135) for BUGFIX-0002.

Companion (§11.4.18) to the in-source documentation block at the top of the script. Pairs with docs/issues/fixed/BUGFIXES.md (BUGFIX-0002).

Overview

A standing regression guard that proves the proxy orchestrator creates the squid **log** bind-mount directory **world-writable**, so the squid container's non-root proxy user (remapped to a high subuid under rootless Podman) can open `/var/log/squid/access.log`. Without it squid FATALs and crash-loops and the proxy serves nothing — the BUGFIX-0002 defect.

It does **not** grep the source. It exercises the **real** orchestrator code: it sources `lib/container-runtime.sh` with the environment pointed at a throwaway temp root, calls `create_directories()`, then stats the resulting log dir and asserts the world-write bit is set.

Prerequisites

- bash (the guard sources the bash-only runtime lib inside an explicit `bash -c` so the outer `/bin/sh` parser sees only a string — §11.4.67).
- GNU stat, mktemp. **Linux-only** — the rootless-Podman bind-mount UID-shift this guards is a Linux `/etc/subuid` mechanism (§11.4.81 honest scope; on a platform without subuid remapping the guard is N/A).

Usage examples

```
# GREEN guard (default) – assert the real orchestrator makes
LOG_DIR writable:
tests/regression/log_dir_writable_test.sh           # exit 0 =
PASS

# RED reproduce – replicate the pre-fix behaviour and assert the
defect:
RED_MODE=1 tests/regression/log_dir_writable_test.sh # exit 0 =
defect reproduced

# Runs automatically inside the suite:
bash tests/run-tests.sh                             #
test_regression_guards()
```

§11.4.115 RED_MODE polarity

RED_MODE	What it runs	PASS means
0 (default)	the REAL create_directories() from the lib	LOG_DIR is world- writable → the fix is present (GREEN guard)
1	the PRE-FIX replica (mkdir only, no chmod)	LOG_DIR is NOT world-writable → the defect reproduces

A RED_MODE=1 run that *cannot* reproduce (dir already writable without the fix) is a finding per §11.4.7, not a pass.

Edge cases

- **Fix reverted / removed** → the GREEN guard reports REGRESSION: ... mode=755 NOT world-writable and exits 1. This is the §1.1 paired-mutation behaviour: stripping chmod 777 "\$LOG_DIR" from create_directories() makes the guard FAIL (proven byte-identical-restore in BUGFIX-0002).
- **Non-Linux host** → stat -c / subuid semantics differ; treat as N/A.
- **Temp dir** is created with mktemp -d and removed on every exit path (trap ... EXIT INT TERM, §11.4.14).

Internal behaviour

- #!/bin/sh, set -eu; POSIX-only constructs in the outer script (bash-only lib sourcing is wrapped in bash -c).

- World-write detection: last octal digit of `stat -c '%a'` has the 2-bit set (2|3|6|7).
- Writes one evidence file per run under `qa-results/regression/bugfix38/` (gitignored).

Related

- Fix sites: `lib/container-runtime.sh` (`create_directories()`), `start` (`init_cache()`).
- `docs/issues/fixed/BUGFIXES.md` — BUGFIX-0002 root-cause + verification.
- `tests/run-tests.sh` — registers this guard via `test_regression_guards()`.

Last verified

2026-07-01 — `sh -n + bash -n parse-clean`; RED reproduces (mode 755), GREEN proves the fix (mode 777); §1.1 mutation makes the GREEN guard FAIL and the lib restores byte-identical (md5 0128a96b6d467c2da5b7cef8a808e563).